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A low-complexity fully digital timing-skew calibration for arbitrary-channel TIADCs

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ABSTRACT

—This paper presents a fully digital background timing-skew calibration method for time-interleaved ADCs (TIADCs) with arbitrary numbers of channels. Compared with existing techniques, the proposed method has four key features. First, a feedforward non-iterative structure enables fast calibration while supporting arbitrary-channel TIADCs. Second, it relies only on statistical information of channel outputs, without requiring reference channels or auxiliary hardware. Third, a closed-form solution is derived from the recursive relationship among timing skews, avoiding matrix inversion and reducing complexity. Fourth, it cancels the impact of FIR filter high-frequency degradation, ensuring robust performance across different Nyquist zones. MATLAB simulation results show that after calibration, the SNDR and SFDR reach 67.54 dB and 94.09 dB, respectively, corresponding to improvements of 11.67 dB and 32.74 dB, with the calibration converging in about 10K samples. These results demonstrate that the proposed method achieves high accuracy, fast convergence, and good robustness, making it suitable for arbitrary-channel and high-speed TIADC systems.

1. Introduction

With the rapid development of broadband wireless communication, radio, and RF sampling receiver systems, analog-to-digital converters (ADCs) play a key role in high-speed data acquisition. Such applications usually require high sampling rate, high resolution, and low power consumption at the same time, while single-channel ADCs have an inherent trade-off between speed and power. Time-interleaved analog-to-digital converters (TIADCs) employ multiple sub-ADCs operating in parallel to increase the overall sampling rate while maintaining controllable per-channel implementation complexity, and have become an important approach for achieving ultra-high-speed data conversion [1]- [2].

However, TIADC performance suffers from channel mismatches including offset, gain and timing mismatches due to process, voltage, and temperature variations [1]. These non-idealities destroy the uniform sampling structure and generate interleaving spurs in the spectrum, significantly degrading the SNDR and SFDR [3], [4]. Among them, the timing-skew error increases with the input signal frequency, making it a key factor limiting system performance [1], [5], [6]. Therefore, accurate estimation and compensation of timing mismatch are critical for calibration.

To address timing mismatch in TIADCs, various calibration methods have been proposed, including analog, mixed-signal, and fully digital approaches. Analog methods adjust sampling clocks or delay units in the analog domain [7], while mixed-signal methods combine analog tuning with digital processing for timing alignment [1], [8], [9], [10]. Fully digital methods rely entirely on digital signal processing to estimate and compensate mismatches [5], [2], [11], [12], [13], [14], [15]. With the advancement of digital circuits, fully digital calibration has become the mainstream solution, enabling online calibration without increasing analog complexity [16].

Nevertheless, existing timing-skew calibration methods still generally suffer from the following problems: (1) slow convergence speed; (2) reliance on additional structures or reference channels, leading to high implementation complexity; (3) restriction to 2^n-channel architectures; (4) limited applicability in high Nyquist zones.

First, LMS-based calibration suffers from a tradeoff between convergence speed and accuracy determined by the step size. Although binary search [14] improves convergence compared to LMS, it remains an iterative method that depends on the initial search range and error monotonicity, requires multiple evaluations, and is not well suited for global multi-channel estimation.

Second, some methods enhance error observability by introducing

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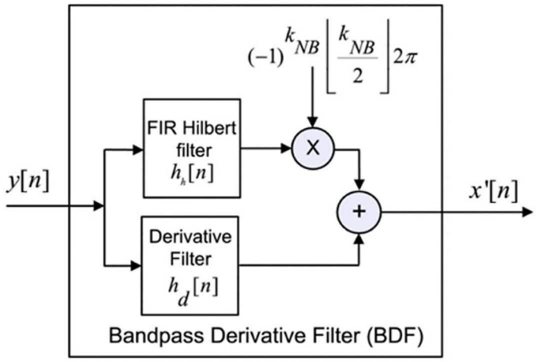


Fig. 1. Bandpass derivative filter (BDF) structure [1].

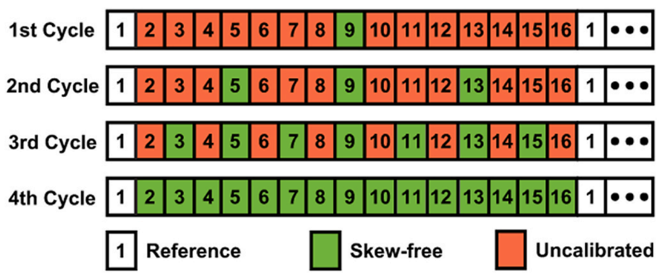


Fig. 2. Binary-grouping calibration method [7].

auxiliary sampling channels or split-based structures. For example, split-ADC and digital-mixing methods extract error information through structural extension or inter-channel correlation differences [17], [18]. Although these methods improve estimation accuracy, they usually require additional hardware, thereby increasing system complexity and power consumption.

Third, conventional timing-skew calibration methods suffer from significant performance degradation in high Nyquist zones. To address this issue, several approaches employ bandpass derivative filters (BDFs) to enable high-frequency calibration [1], [5], [3], [2], [19], whose structure is shown in Fig. 1. However, BDF-based methods go beyond simple FIR differentiators, as they typically incorporate Hilbert transform and bandpass filtering structures, resulting in increased implementation complexity and hardware overhead.

Fourth, many existing calibration methods lack scalability to arbitrary-channel TIADCs, as they rely on specific structures that are not applicable to non-2^n channel configurations [7]. For example, as shown in Fig. 2, the digital-mixing-based method in Ref. [7] adopts a hierarchical calibration scheme where channels are calibrated in a binary grouping manner, leading to a structure that inherently favors 2^n-channel systems. Although matrix-based methods [1] are more general, they require matrix inversion, resulting in high computational complexity and significant hardware overhead, especially as the number of channels increases.

Recently, intelligent calibration techniques have also been introduced to improve timing-skew estimation in TIADCs. Reference [20] proposed a frequency-fitness genetic algorithm (FFGA), which estimates the timing skew of each channel independently and supports arbitrary interleaving factors. Reference [21] further adopted fast adaptive particle swarm optimization (FAPSO) to realize simultaneous channel convergence and fast background calibration. In addition, a frequency-recognition CNN (FR-CNN) method was reported to improve calibration accuracy for different input-frequency ranges and signal types [22]. Although these methods achieve good calibration performance and accurate estimation of skews, they generally rely on iterative optimization or neural-network training, which may introduce additional convergence time, hardware resources, or implementation complexity.

To address the above limitations, this paper proposes a fully digital timing-skew calibration method applicable to TIADCs with an arbitrary number of channels. The proposed method adopts a feedforward, non-iterative structure to achieve fast timing-skew estimation. It relies solely on the statistical information of existing channel outputs, without requiring additional hardware or structural modification, thereby reducing implementation complexity. The effect of FIR-based derivative degradation at high frequencies is effectively canceled, alleviating bandwidth limitations and maintaining robust performance across different Nyquist zones. Furthermore, a recursive timing-skew model with a closed-form solution is derived, transforming the calibration into a low-complexity computation and avoiding matrix inversion. As a result, the proposed method achieves fast convergence, low complexity, high scalability, and robust performance across different Nyquist zones, while naturally supporting arbitrary-channel configurations.

The remainder of this paper is organized as follows. Section II presents the timing-mismatch model and the proposed recursive closed-form calibration method. Section III provides MATLAB simulation results. Section IV concludes the paper.

2. System model and proposed method

2.1. Mathematical model

Under ideal conditions, each channel samples the input signal x(t) alternately with a uniform sampling interval T_s. However, in practical systems, due to mismatches in the clock distribution paths, the sampling instants of each channel deviate, resulting in asymmetric perturbations of the sampling intervals [1], [5].

For an M-channel TIADC, the actual sampling instant of the i-th sub-ADC ($t_i[n]$) can be expressed as:

$$t_i[n] = nMT_s + iT_s + \delta_i \tag{1.}$$

where the first two terms represent the ideal uniform sampling instants, while δ_i represents the timing skew of the i-th channel and M is the number of channels.

To describe the impact of timing mismatch on the statistical characteristics of the system, the autocorrelation function of the input signal is introduced:

$$R_x(\tau) = E\{x(t)x(t+\tau)\} \tag{2.}$$

where $E\{\cdot\}$ denotes the expectation operator, and τ is the time lag. $R_x(\tau)$ measures the similarity between the signal and its delayed version. Based on this statistical model, existing studies have shown that the cross-correlation between channels can be used to characterize the impact of timing skew on the system output [2], [7], [23].

For this purpose, the correlation difference between adjacent channels is introduced:

$$\Gamma_i = R_{y_i y_{i-1}}(0) - R_{y_i y_{i+1}}(0) \tag{3.}$$

Under the small-skew condition $|\delta_i| \ll T_s$, an approximate linear relationship between Γ_i and the timing skew can be obtained [1], [5]:

$$\Gamma_i \approx R_x'(T_s)[(\delta_i - \delta_{i-1}) - (\delta_{i+1} - \delta_i)] \tag{4.}$$

It should be noted that (4) is based on the small-skew assumption. For large timing skews, higher-order Taylor terms may introduce model mismatch, making the proposed method more suitable for fine timing-skew calibration after clock-path matching.

The above results show that the timing skews among channels satisfy a recursive structure, and the estimation problem can be transformed into a discrete recursive problem driven by correlation differences, which provides the basis for the design of subsequent calibration algorithms.

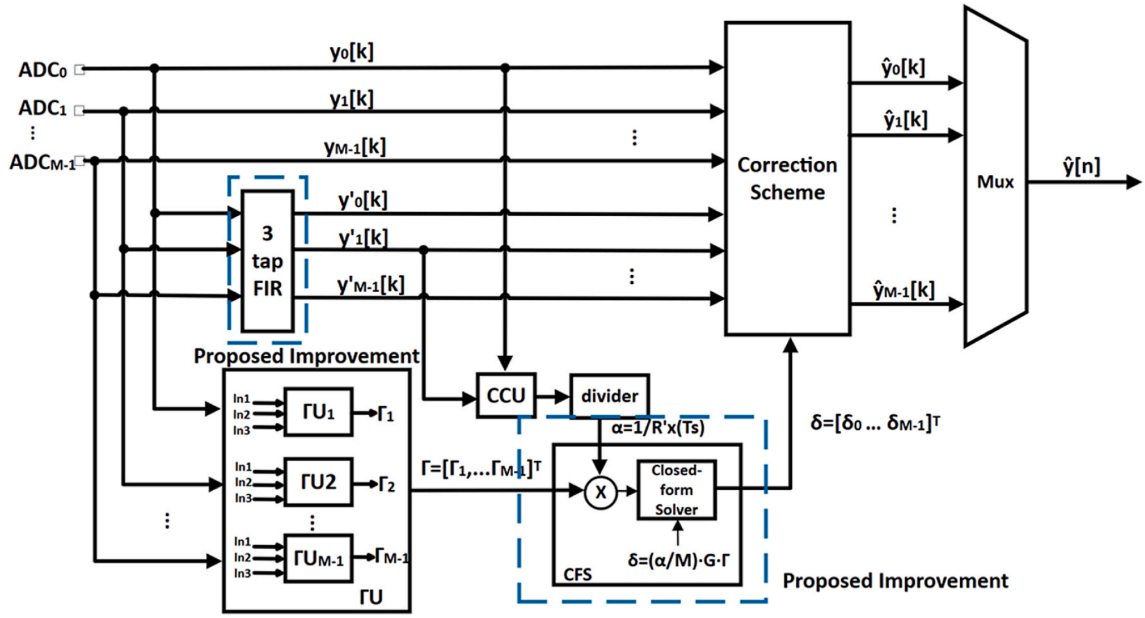


Fig. 3. The overall architecture of the proposed calibration.

2.2. Limitations of conventional matrix-based calibration method

By modeling the inter-channel correlation statistics, a linear relationship between the timing skew and the observations can be obtained [1]:

$$\boldsymbol{\Gamma} = \frac{1}{R_x'(T_s)}\mathbf{H}\cdot\boldsymbol{\delta} \tag{5.}$$

where $\boldsymbol{\delta}$ is the timing-skew vector of all channels, $\boldsymbol{\Gamma}$ is the statistical quantity constructed from the cross-correlation functions, and matrix $\mathbf{H}$ describes the mapping relationship between them. Existing studies have shown that this matrix has a tridiagonal symmetric structure [1], and its typical form is:

$$\mathbf{H} = \begin{pmatrix} 2 & -1 & 0 & \dots & -1 \\ -1 & 2 & -1 & \dots & 0 \\ 0 & -1 & 2 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -1 & 0 & 0 & -1 & 2 \end{pmatrix}_{M\times M} \tag{6.}$$

Matrix $\mathbf{H}$ is a circulant matrix, and the sum of the elements in each row and each column is zero. Therefore, its rank is M-1. Specifically, as shown in Ref. [1], the summation of all equations is independent of the timing skew variables, indicating that the linear equation system contains redundant constraints.

Therefore, the original equation system is an underdetermined system and cannot directly yield a unique solution. To address this issue, one row of the matrix is usually removed to transform it into a full-rank form. Specifically, by deleting the first row of matrix $\mathbf{H}$, a new matrix $\mathbf{A}$ is constructed, thereby obtaining a full-rank linear equation system and enabling effective estimation of the timing skews [1].

After removing the first row, the above relationship can be simplified as:

$$\boldsymbol{\Gamma}_{\mathbf{tail}} = \frac{1}{R_x'(T_s)}\mathbf{A}\cdot\boldsymbol{\delta}_{\mathbf{tail}} \tag{7.}$$

where matrix $\mathbf{A}$ is:

$$\mathbf{A} = \begin{pmatrix} -1 & 2 & -1 & \dots & 0 \\ 0 & -1 & 2 & \dots & 0 \\ 0 & 0 & -1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -1 & 0 & 0 & -1 & 2 \end{pmatrix}_{(M-1)\times M} \tag{8.}$$

Since this linear system is generally not directly invertible, it needs to be solved using the least-squares methods:

$$\boldsymbol{\delta}_{\mathbf{T}} = \begin{pmatrix} \boldsymbol{\delta}_{\mathbf{t0}} \\ \boldsymbol{\delta}_{\mathbf{t1}} \\ \vdots \\ \boldsymbol{\delta}_{\mathbf{t(M-1)}} \end{pmatrix} \approx \mathbf{A}^{\mathrm{T}}\left(\mathbf{A}\mathbf{A}^{\mathrm{T}}\right)^{-1}\frac{1}{R_x'(T_s)}\begin{pmatrix} \boldsymbol{\Gamma}_{\mathbf{1}} \\ \boldsymbol{\Gamma}_{\mathbf{2}} \\ \vdots \\ \boldsymbol{\Gamma}_{\mathbf{M-1}} \end{pmatrix} \tag{9.}$$

where $\boldsymbol{\delta}_{\mathbf{T}}$ is an M×1 timing-skew vector, $\boldsymbol{\Gamma}$ is an (M − 1)×1 statistical vector, and $\mathbf{A}$ is an (M − 1)×M matrix obtained by removing the first row of $\mathbf{H}$. Moreover, $\mathbf{A}$ is full-rank, and thus $\mathbf{A}\mathbf{A}^{\mathbf{T}}$ is invertible.

Although the above method can estimate the timing skews from a global perspective, it still has several limitations in practical applications.

(1) **High computational complexity.** The solution requires matrix multiplication and inversion, typically with complexity on the order of cubic scaling, which significantly limits real-time capability in high-dimensional cases.
(2) **Large hardware overhead.** Matrix inversion requires substantial additional computational and hardware overhead due to its complex operations.
(3) **Underutilization of structure.** Although timing skews follow an inherent recursive relationship, matrix-based methods treat them as globally coupled variables, introducing unnecessary computational redundancy.

In summary, conventional matrix-based timing-skew calibration methods suffer from limitations in computational complexity, hardware overhead, and inefficient structural utilization. Therefore, it is necessary to develop a low-complexity calibration method that directly exploits the difference structure, avoids matrix inversion, supports an arbitrary number of channels, and remains effective in high Nyquist zones.

2.3. Proposed calibration algorithm

As shown in Fig. 3, the proposed calibration architecture consists of a 3-tap FIR differentiator, a Gamma Unit ($\boldsymbol{\Gamma}$U) based error extraction module, a scaling factor estimation block (CCU and divider), and a closed-form solver (main improvement), followed by a digital correction scheme and MUX reconstruction. Specifically, the sub-ADC outputs $y_i[n]$ are first processed by the 3-tap FIR filter to obtain their discrete-time derivatives $y_i'[n]$, which reflect the sensitivity of the signal to timing perturbations. The $\boldsymbol{\Gamma}$U module is responsible for constructing the error vector $\boldsymbol{\Gamma}$ by exploiting the statistical relationship between adjacent channel outputs, thereby generating the correlation-based error terms Γ_i. Meanwhile, the CCU and divider estimate the normalization factor

$$\alpha = \frac{1}{R_x'(T_s)} \tag{10.}$$

where the $R_x'(T_s)$ can be calculated by:

$$\begin{aligned} R_x'(T_s) &= E\left\{ \underbrace{x'(nMT_s + (i+1)T_s + t_0)}_{x_{i+1}'[n]} \cdot \underbrace{x(nMT_s + iT_s + t_0)}_{x_i[n]} \right\} \\ &= E\{x_{i+1}'[n] \cdot x_i[n]\} \end{aligned} \tag{11.}$$

Based on Γ_i and α, the closed-form solver directly computes the timing skews δ_i, which are subsequently applied in the correction block to compensate each channel output before recombination through the MUX.

From the correlation-based modeling, the timing skews satisfy the equation

$$\delta_{i+1} = 2\delta_i - \delta_{i-1} - \alpha\Gamma_i \tag{12.}$$

Since only relative timing offsets are identifiable, the first channel is selected as the reference, and the initial conditions are defined as $\delta_1 = 0$, $\delta_2 = s$.

To illustrate the operation of the proposed method, a five-channel TIADC is first considered. Starting from the initial conditions, the recursive relationship yields

$$\delta_3 = 2s - \alpha\Gamma_2 \; \delta_4 = 3s - 2\alpha\Gamma_2 - \alpha\Gamma_3 \; \delta_5 = 4s - 3\alpha\Gamma_2 - 2\alpha\Gamma_3 - \alpha\Gamma_4 \tag{13.}$$

Considering the periodic structure of the interleaved system, the boundary condition is given by $\delta_6 = \delta_1 = 0$. From this condition, the unknown parameter can be obtained as

$$s = \frac{\alpha}{5}(4\Gamma_2 + 3\Gamma_3 + 2\Gamma_4 + \Gamma_5) \tag{14.}$$

Thus, the timing skews of all channels can be explicitly determined without iterative procedures.

By examining the expressions of the first few channels, it can be observed that the coefficient of s is i-1, while the coefficients of Γ_j decrease linearly. Based on this pattern, the general expression of the timing skew for an arbitrary channel can be written as

$$\delta_i = (i-1)s - \alpha\sum_{j=2}^{i-1}(i-j)\Gamma_j \tag{15.}$$

To determine the unknown parameter s, the periodic condition is applied as $\delta_{M+1} = \delta_1 = 0$, and thus

$$s = \frac{\alpha}{M}\sum_{j=2}^{M}(M+1-j)\Gamma_j \tag{16.}$$

Substituting this result into the general expression, the closed-form solution of timing skews for all channels can be obtained directly.

For implementation convenience, the above formulation can be further expressed in matrix form as

$$\boldsymbol{\delta} = \frac{\alpha}{M}\mathbf{G}\boldsymbol{\Gamma} \tag{17.}$$

For the five-channel case, the corresponding matrix is:

$$\mathbf{G} = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 3 & 6 & 4 & 2 \\ 2 & 4 & 6 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix} \tag{18.}$$

Extending to M channels, the matrix elements are defined as

$$\mathbf{G_{r,c}} = \min(ind_{row}, ind_{col})(M - \max(ind_{row}, ind_{col})), ind_{row}, ind_{col} = 1, \ldots, M - 1 \tag{19.}$$

where ind_{row} and ind_{col} denote the row and column indices of matrix $\mathbf{G}$.

It should be emphasized that this matrix is directly constructed from the recursive difference relationship rather than obtained through matrix inversion, thereby significantly reducing computational complexity and hardware overhead.

The obtained $\boldsymbol{\delta}$ is an underdetermined solution. Based on physical considerations, a zero-mean condition is imposed on the estimated skews, which removes the average timing skew across all channels:

$$\widetilde{\delta}_i = \delta_i - \frac{1}{M}\sum_{j=1}^{M}\delta_j \tag{20.}$$

where $\widetilde{\delta}_i$ represents the zero-mean timing skew.

After estimating the timing skews, each channel is corrected as

$$\widehat{y}_i[n] = y_i[n] - \widetilde{\delta}_i x_i'[n] \tag{21.}$$

where $y_i[n]$ is the raw output of the i-th channel, $\widehat{y}_i[n]$ is the calibrated output, and $\widehat{x_i'}[n]$ is the derivative of the input signal obtained by a 3-tap FIR differentiator.

Due to the non-ideal frequency response of the differentiator [14], the estimated derivative can be approximated as a scaled version of the ideal derivative:

$$\widehat{x}_{i+1}'[n] = \beta_k x_{i+1}'[n] \tag{22.}$$

where β_k is a scaling factor dependent on both the number of taps and the signal frequency. Meanwhile, based on (11),

$$\begin{aligned} \widehat{R_x'}(T_s) &= E\{\widehat{x_{i+1}'}[n] \cdot x_i[n]\} \\ &= \beta_k E\{x_{i+1}'[n] \cdot x_i[n]\} \\ &= \beta_k R_x'(T_s) \end{aligned} \tag{23.}$$

Furthermore, the factor α is proportional to the inverse of the derivative of the autocorrelation function. Therefore, it is scaled by $1/\beta_k$, yielding the estimated coefficient $\widehat{\alpha}$.

$$\widehat{\alpha} = \frac{1}{\widehat{R_x'}(T_s)} = \frac{1}{\beta_k} \cdot \frac{1}{R_x'(T_s)} = \frac{1}{\beta_k}\alpha \tag{24.}$$

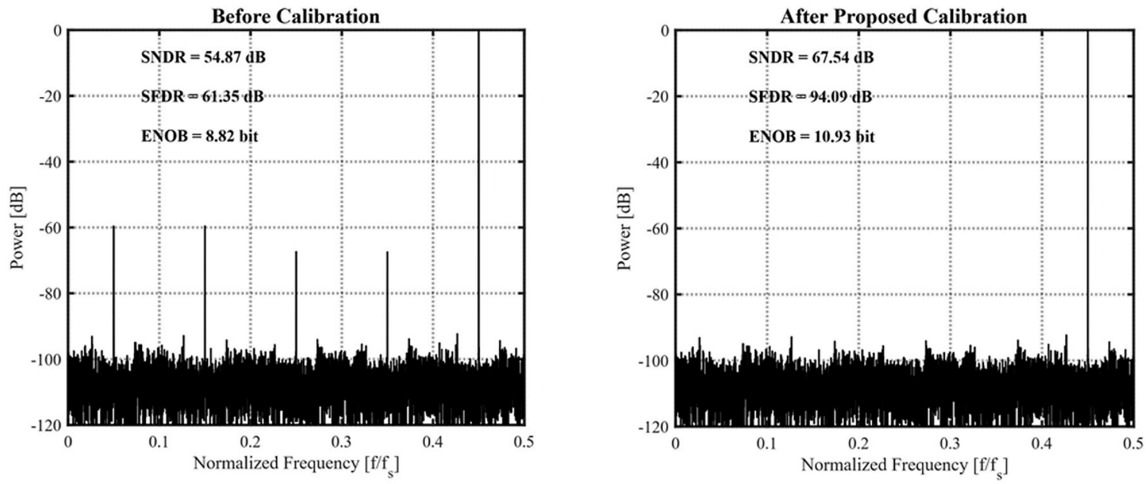
From (17), $\boldsymbol{\delta}$ is proportional to $\boldsymbol{\alpha}$. Thus, we obtain

$$\widehat{\boldsymbol{\delta}} = \frac{\widehat{\alpha}}{M}\mathbf{G}\boldsymbol{\Gamma} = \frac{1}{\beta_k}\frac{\alpha}{M}\mathbf{G}\boldsymbol{\Gamma} = \frac{1}{\beta_k}\boldsymbol{\delta} \tag{25.}$$

According to (22) and (25), in the final compensation term, the scaling effects cancel each other, yielding

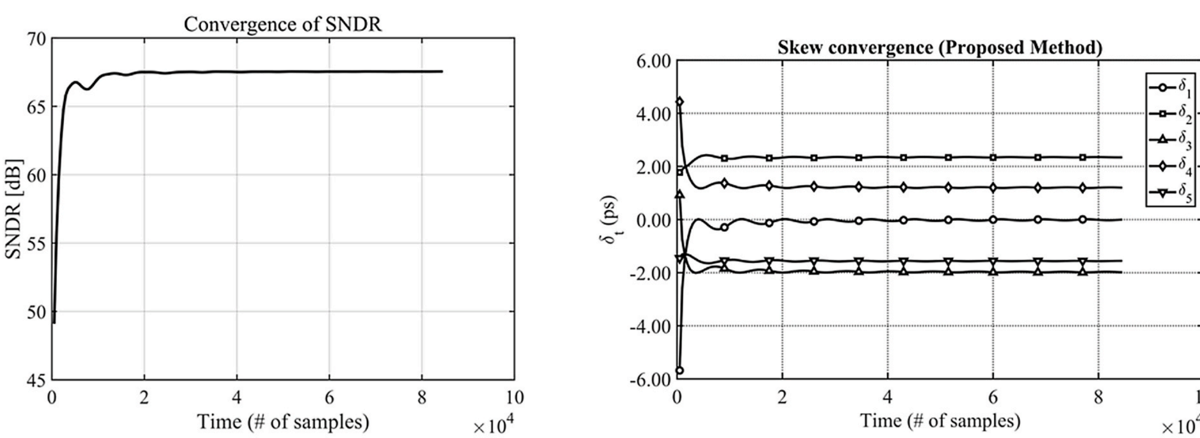
$$\begin{aligned} \widehat{y}_i[n] &= y_i[n] - \widehat{\delta}_i \widehat{x_i'}[n] \\ &= y_i[n] - \frac{1}{\beta_k}\delta_i \cdot \beta_k x_i'[n] \\ &= y_i[n] - \delta_i x_i'[n] \end{aligned} \tag{26.}$$

This indicates that the influence of the tap number is largely



(a) (b)

Fig. 4. Output spectra of the TIADC before and after calibration under single-tone input at frequency $f_{in} = 0.45f_{s,total}$: (a) Before calibration; (b) After proposed calibration.



(a) (b)

Fig. 5. Convergence behavior of the proposed method at frequency $f_{in} = 0.45f_{s,total}$: (a) SNDR convergence; (b) timing-skew estimation convergence.

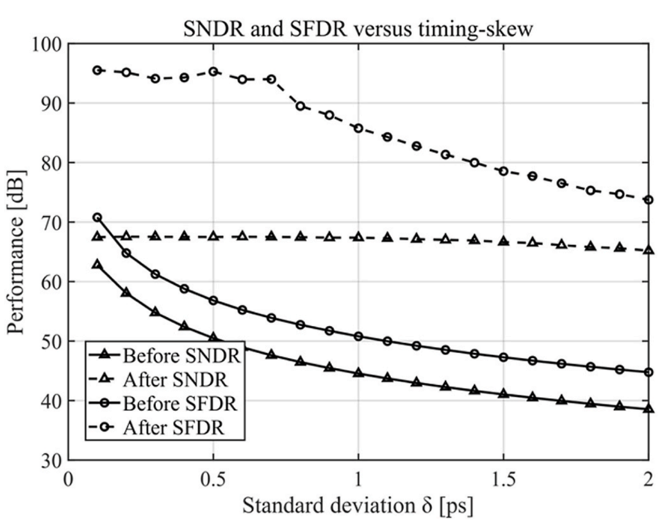


Fig. 6. SNDR and SFDR performance versus standard deviation δ from 0.1 ps to 2 ps at frequency $f_{in} = 0.45f_{s,total}$.

suppressed in the final calibration result, making the proposed method insensitive to the differentiator length. As a result, a simple 3-tap FIR filter is sufficient for accurate calibration.

3. Simulation results

To verify the effectiveness of the proposed method, a TIADC system-level simulation model is built on the MATLAB platform. The system adopts a 5-channel TIADC architecture, with a total sampling rate of 2.7 GS/s (540 MS/s per channel). The ADC resolution is 11 bits, with a reference voltage of 0.9V, a full-scale voltage of 0.88V, and a common-mode voltage of 0.45V. The input signal is a single-tone sinusoidal signal, whose frequency is set as $f_{in} = 0.45f_{s,total}$, where $f_{s,total}$ is the total sampling rate. To ensure the stability and reproducibility of the simulation results, a fixed timing-skew vector $[0, 0.33, -0.28, 0.17, -0.22]$ps is adopted, where the first channel is used as the reference channel.

In the baseline simulation, timing-skew mismatch is first considered to independently evaluate the proposed timing-skew calibration. To further verify the robustness under more practical TIADC conditions, additional simulations including gain and offset mismatches are also performed. The total number of samples is set to 85000, and the FFT size is 16384.

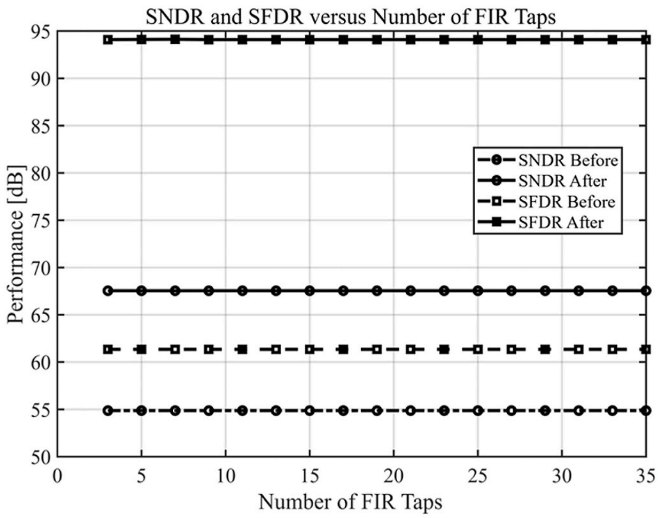


Fig. 7. SNDR and SFDR versus number of FIR taps at frequency $f_{in} = 0.45f_{s,total}$.

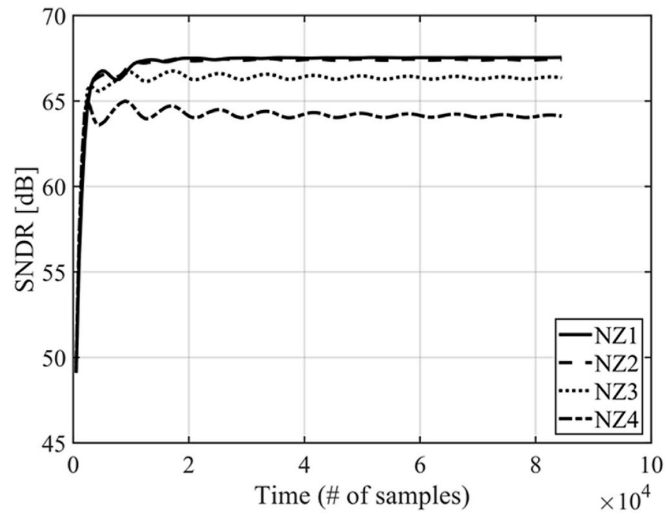


Fig. 8. SNDR convergence behavior under different Nyquist zones (NZ1–NZ4) at a frequency of $0.45 \times fs + (k_{NB} - 1) \times fs$, $k_{NB} \in \{1, 2, 3, 4\}$.

Fig. 4 presents the output spectra of the TIADC before and after calibration under a single-tone input at frequency $f_{in} = 0.45f_{s,total}$, with a fixed timing-skew vector $[0,0.33, -0.28,0.17, -0.22]$ps. As observed in Fig. 4(a), the presence of timing mismatch introduces evident spurious components, which degrade the spectral performance. With the proposed calibration applied, Fig. 4(b) shows that these spurious tones are largely eliminated. The SFDR is enhanced by approximately 32.74 dB, while the SNDR improves to 67.54 dB, reaching a level close to the ideal case.

Fig. 5 shows the convergence behavior of the proposed calibration method at frequency $f_{in} = 0.45f_{s,total}$, with a fixed timing-skew vector [0, 0.33, −0.28, 0.17, −0.22]ps. As illustrated in Fig. 5(a), the SNDR increases rapidly from about 50 dB to approximately 66 dB within the first 0.5×10^4 samples, and then gradually converges to around 67 dB after about 1×10^4 samples. Fig. 5(b) presents the convergence of the timing-skew estimates for all channels. The estimated skews approach their steady-state values within approximately 1×10^4 samples, corresponding to about 3.7 μs at a sampling rate of 2.7 GS/s, with only small residual fluctuations. The convergence speed is influenced by factors such as input frequency, number of channels, and mismatch level.

Fig. 6 shows the SNDR and SFDR performance versus timing-skew at frequency $f_{in} = 0.45f_{s,total}$. Timing skews are modeled as Gaussian distribution with zero mean and standard deviation δ of 0.1ps to 2ps. Without calibration, both SNDR and SFDR degrade significantly as the timing-skew increases. Specifically, the SNDR decreases from about 63 dB to 39 dB, while the SFDR drops from around 71 dB to approximately 45 dB as the skew increases from 0.1 ps to 2 ps, which is 0.027% Ts to 0.54%Ts. After applying the proposed calibration, the performance is substantially improved across the entire skew range. The SNDR is improved to about 68 dB, while the SFDR is enhanced to about 95 dB at 0.1 ps and remains around 75 dB when the skew approaches 2 ps. These results indicate that the proposed method can effectively suppress timing-skew-induced degradation over an enlarged mismatch range. However, the calibrated SNDR and SFDR gradually decrease as the timing-skew increases, which is consistent with the limitation of the first-order approximation in (4). Therefore, coarse timing alignment or second-order compensation may be required for extremely large timing-skew cases.

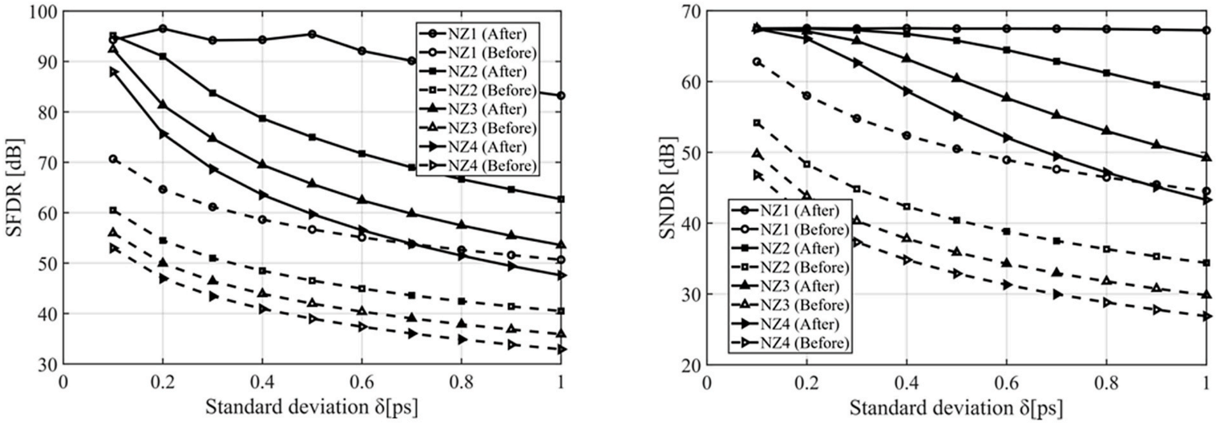
The impact of the FIR tap number on the calibration performance is illustrated in Fig. 7. The input signal is at frequency $f_{in} = 0.45f_{s,total}$, with a fixed timing-skew vector $[0,0.33, -0.28,0.17, -0.22]$ps. It can be observed that both SNDR and SFDR remain nearly invariant with respect to the number of taps over the evaluated range. Before calibration, the SNDR is approximately 55 dB, while after calibration it consistently improves to about 67.5 dB, corresponding to an enhancement of roughly 12.5 dB. Similarly, the SFDR increases from about 61 dB to around 94 dB, achieving an improvement exceeding 30 dB.

Unlike conventional approaches where performance strongly depends on the FIR length, the proposed method exhibits negligible sensitivity to the number of taps. This indicates that even with a small number of taps (e.g., 3 taps), the calibration can already achieve near-optimal performance, and increasing the tap count does not lead to further noticeable improvement.

First, in terms of convergence behavior for the single-tone sinusoid input at a frequency of $0.45f_s + (k_{NB} - 1) \times f_s$, $k_{NB} \in \{1, 2, 3, 4\}$ with a fixed timing-skew vector [0,0.33,-0.28,0.17,-0.22]ps, as shown in Fig. 8, under all four Nyquist-zone conditions, the proposed method is able to converge within approximately 10^4 samples, demonstrating fast convergence characteristics. At the same time, it can be observed that as the Nyquist-zone index increases (NZ1 → NZ4), the converged SNDR and SFDR decrease slightly. This is because the timing error term is proportional to the derivative of the input signal, and its impact increases with the actual input frequency. However, even under the highest Nyquist-zone condition (NZ4), the proposed method can still maintain relatively high dynamic performance, indicating good stability in high-frequency scenarios.

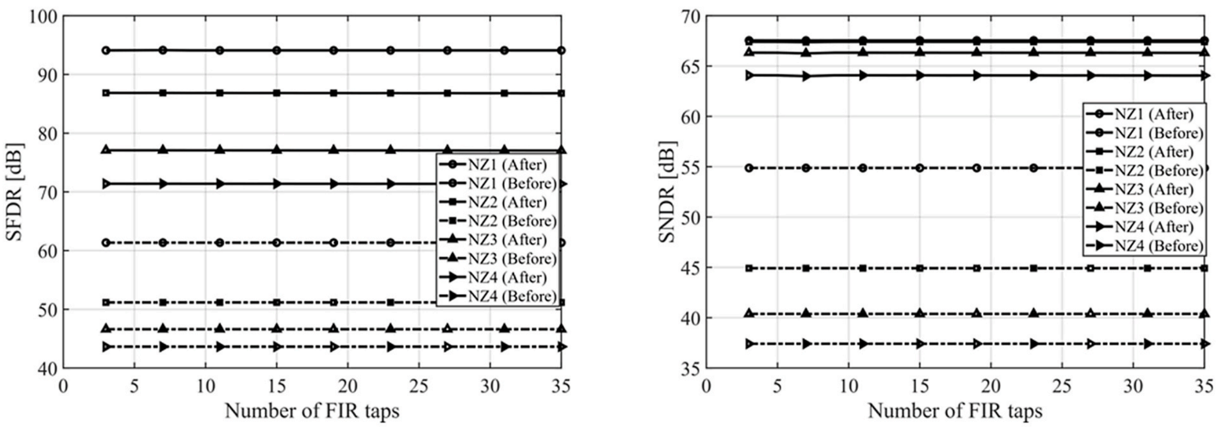
Fig. 9 shows the SNDR and SFDR versus timing-skew across four Nyquist zones (NZ1–NZ4). The solid and dashed curves represent the calibrated and uncalibrated cases, respectively. As the skew increases, both SNDR and SFDR degrade noticeably without calibration, with more severe deterioration observed in higher Nyquist zones. With the proposed calibration, the performance is significantly improved. The SFDR is enhanced by more than 15 dB across all zones, while the SNDR improvement reaches approximately 15–20 dB at large skew levels. In NZ1, the SNDR remains around 67 dB and the SFDR stays above 85 dB over the entire skew range. These results demonstrate that the proposed method effectively suppresses timing-skew-induced distortion and maintains robust performance across different Nyquist zones.

Fig. 10 shows SNDR and SFDR versus the number of FIR taps across NZ1–NZ4. The solid and dash-dot curves represent the calibrated and uncalibrated cases, respectively. Both SNDR and SFDR remain almost unchanged as the tap number increases. For instance, in NZ1, the SNDR is improved from about 55 dB to 67 dB, while the SFDR increases from about 61 dB to 95 dB after calibration. Similar gains are observed in higher Nyquist zones. This indicates that the proposed method is insensitive to the number of FIR taps, and near-optimal performance can be achieved even with a small tap count.



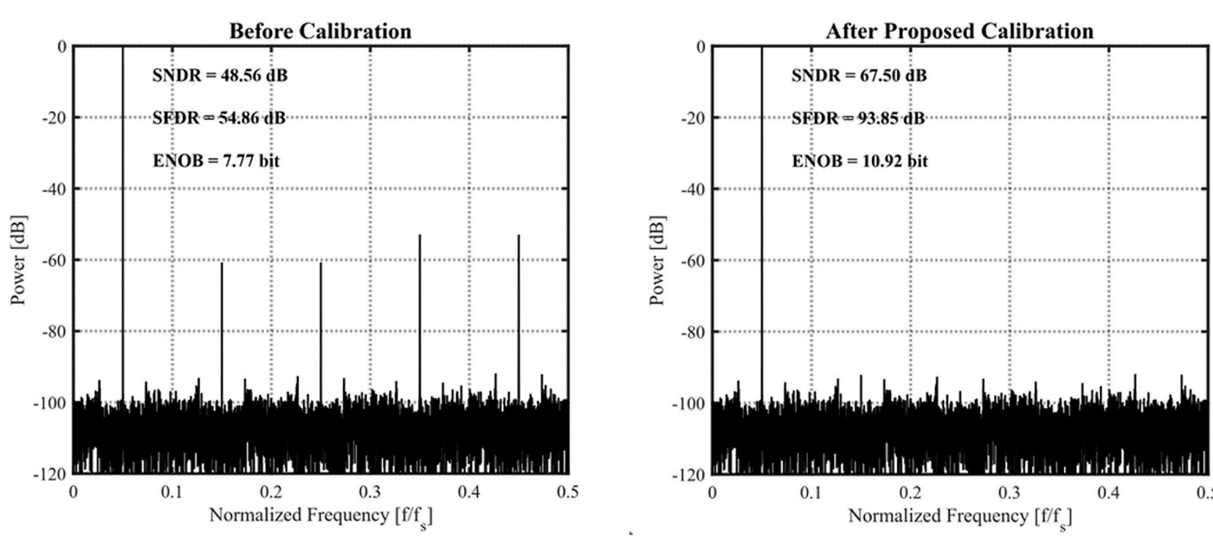
(a) (b)

Fig. 9. SNDR and SFDR versus timing-skew under different Nyquist-zone conditions (NZ1–NZ4) at a frequency of $0.45 \times fs + (k_{NB} - 1) \times fs$, $k_{NB} \in \{1, 2, 3, 4\}$: (a) SFDR; (b) SNDR.



(a) (b)

Fig. 10. SNDR and SFDR versus number of FIR taps under different Nyquist-zone conditions (NZ1–NZ4) at a frequency of $0.45 \times fs + (k_{NB} - 1) \times fs$, $k_{NB} \in \{1, 2, 3, 4\}$: (a) SNDR; (b) SFDR.



(a) (b)

Fig. 11. Output spectra in the second Nyquist zone (NZ2): (a) before calibration; (b) after the proposed calibration.

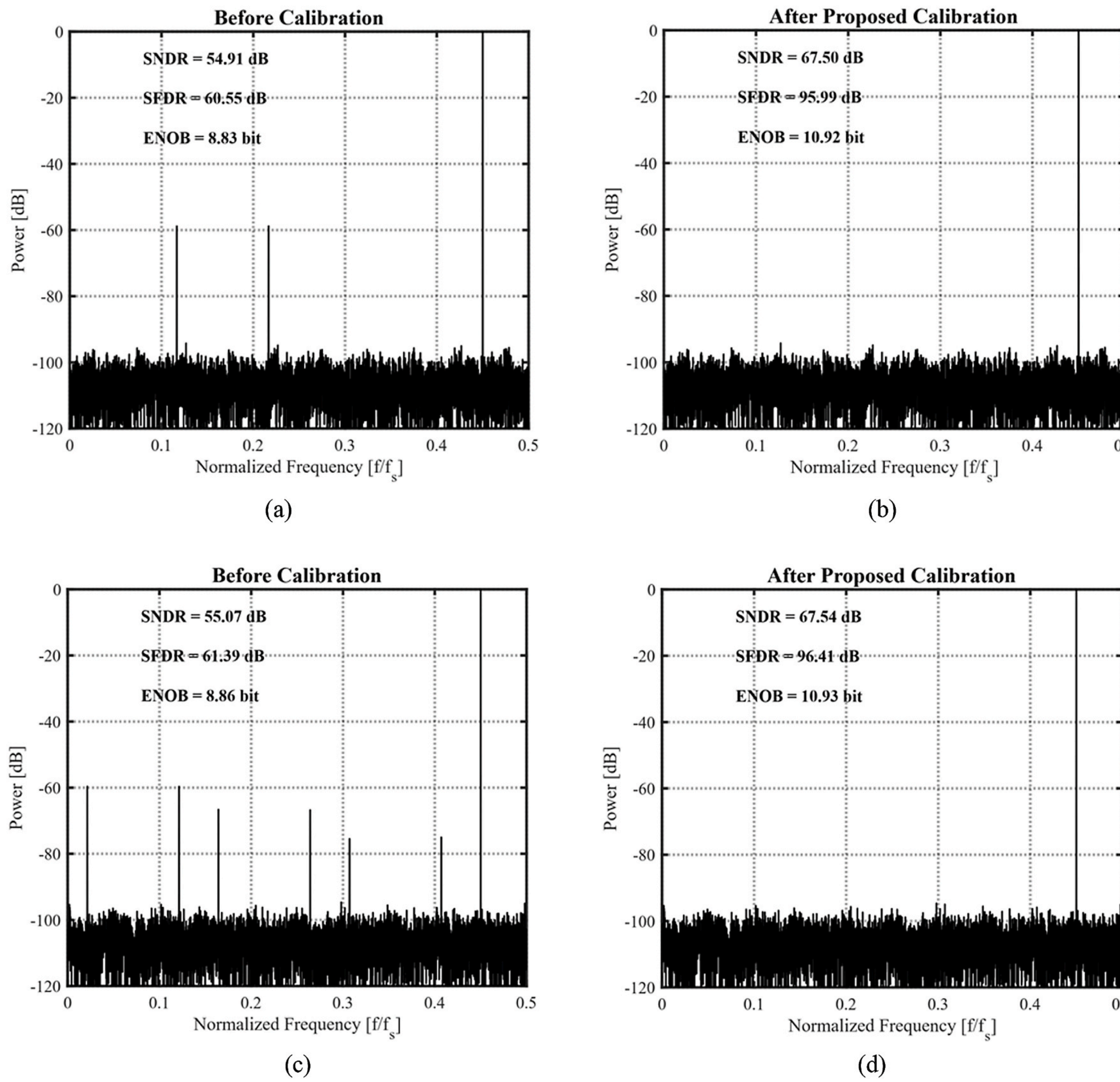


Fig. 12. Output spectra for non-2^n-channel TIADCs at frequency $f_{in} = 0.45f_{s,total}$:(a) 3-channel before calibration; (b) 3-channel after calibration; (c) 7-channel before calibration; (d) 7-channel after calibration.

The case with the input signal located in the second Nyquist zone is selected for analysis, and the spectrum is illustrated by taking the input frequency $f_{in} = 0.45f_{s,total}+0.5f_{s,total}$ as an example.

Fig. 11 shows the output spectra in the second Nyquist zone before and after calibration. The input signal is at frequency $f_{in} = 0.95f_{s,total}$, with a fixed timing-skew vector [0,0.33, −0.28,0.17, −0.22]ps. Before calibration, significant spurious tones caused by timing-skew mismatch are observed, resulting in an SNDR of 48.56 dB and an SFDR of 54.86 dB. After applying the proposed method, the spurious components are effectively suppressed. Consequently, the SNDR improves to 67.50 dB and the SFDR increases to 93.85 dB, corresponding to enhancements of about 19 dB and 39 dB, respectively. This demonstrates the effectiveness of the proposed calibration in high-frequency operation.

To further verify the adaptability of the proposed method to non-2^n channel configurations, additional simulations are performed for 3-channel and 7-channel TIADCs. As shown in Fig. 12, obvious timing-skew-induced spurs appear before calibration in both cases. After applying the proposed calibration, these spurs are effectively suppressed for both the 3-channel and 7-channel configurations. These results confirm that the proposed recursive closed-form solution is not limited to 2^n-channel architectures and can be directly applied to general M-channel TIADC systems.

To verify the robustness of the proposed method under more practical TIADC conditions, gain and offset mismatches are introduced into the behavioral model. The input signal is at frequency $f_{in} = 0.95f_{s,total}$, with a fixed timing-skew vector [0,0.33, −0.28,0.17, −0.22]ps. As shown in Fig. 13, before calibration, the combined mismatches introduce obvious spurious components and degrade the dynamic performance. After gain and offset calibration, the spurs caused by static channel mismatches are suppressed, but residual timing-skew-induced spurs still remain. After applying the proposed timing-skew calibration, these residual spurs are further reduced, and the SNDR and SFDR are improved. These results indicate that the proposed method can still operate effectively in the presence of gain and offset mismatches when combined with conventional gain/offset calibration.

As shown in Table 1, the resource utilization is estimated with respect to a Pango Logos-2 PG2L50H FPGA, which provides 35,800 LUT6s, 71,600 FFs, and 120 APMs. Based on the fixed-point datapath and arithmetic-operator count, the proposed five-channel fully parallel calibration core is estimated to consume 1890 LUTs, 1130 FFs, and 15 DSP/APM multipliers, corresponding to 5.28%, 1.58%, and 12.50% of the available resources, respectively.

Table II compares the proposed method with existing timing-skew calibration techniques. Regarding convergence, the proposed method using feedforward structure requires about 10K samples, which is significantly faster than [24] (24.5K) while maintaining comparable

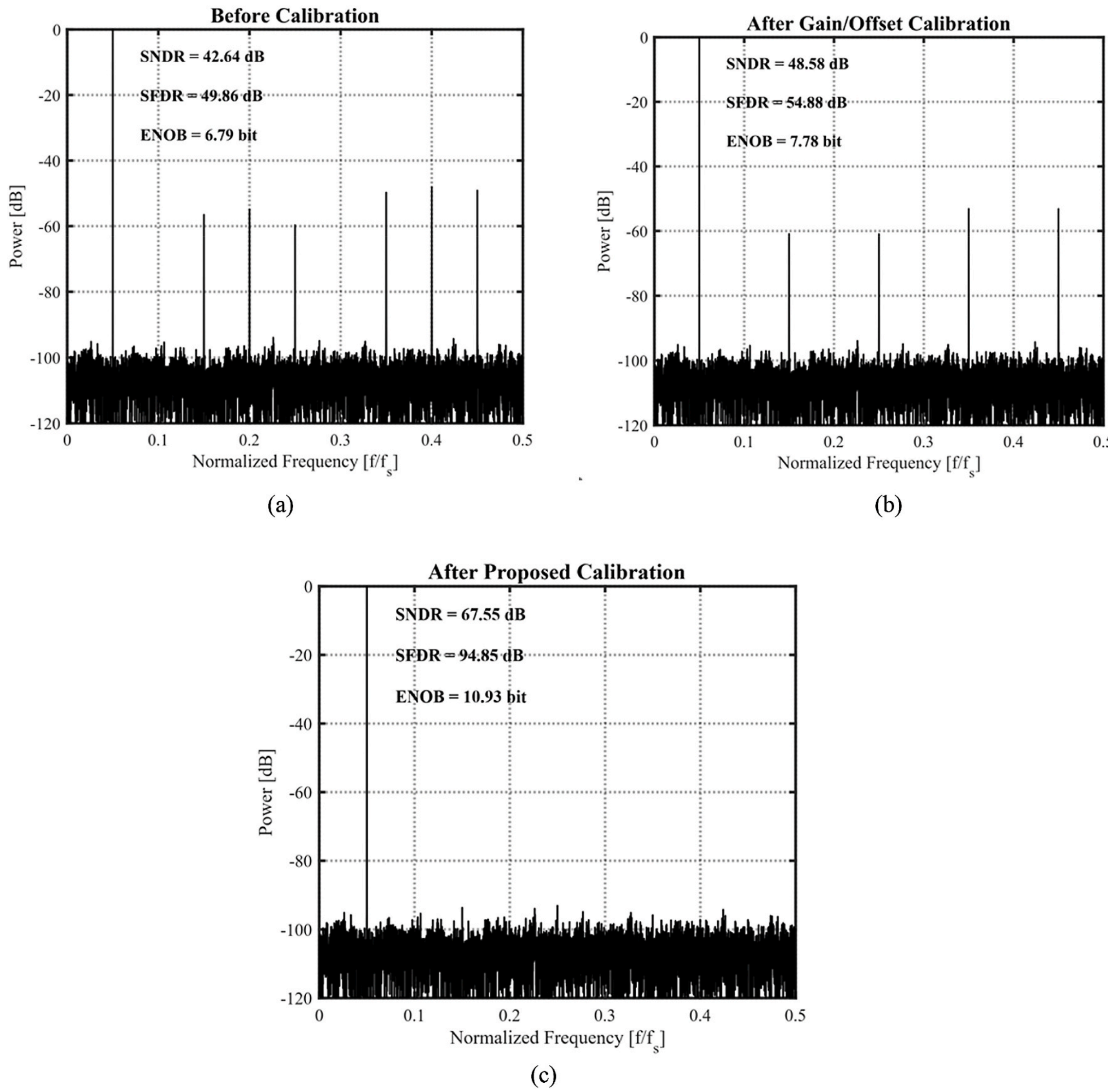


Fig. 13. Output spectra under gain, offset, and timing mismatches at frequency $f_{in} = 0.45f_{s,total}+0.5f_{s,total}$:(a) before calibration; (b) after gain and offset calibration; (c) after gain, offset, and proposed timing-skew calibration.

Table 1
Estimated FPGA resource consumption.

Resource	Estimated consumption	Available on PG2L50H	Utilization
LUTs	1890	35800	5.28%
FFs	1130	71600	1.58%
DSPs/APMs	15	120	12.50%

speed to Refs. [3], [2], and [7]. In terms of implementation complexity, unlike high-order FIR structures in Refs. [2] and [24] or auxiliary ADCs in Ref. [7], the proposed method only employs a 3-tap FIR with neither reference channels nor matrix inversion, resulting in a simpler and more hardware-efficient design. It also supports arbitrary channel configurations, unlike most existing methods [3], [2], [7], [24], and maintains robust performance across multiple Nyquist zones, similar to Refs. [1], [3], [2]. Overall, the proposed method provides high accuracy, rapid convergence, and strong robustness, making it well suited for arbitrary-channel and high-speed TIADC systems.

4. Conclusion

This paper presents a fully digital timing-skew calibration method for TIADCs with an arbitrary number of channels. By exploiting the recursive relationship among channel skews and deriving a closed-form solution, the proposed method avoids matrix inversion and significantly reduces computational complexity. It requires neither reference channels nor auxiliary sampling, and achieves fast estimation without iterative adaptation, resulting in low hardware overhead. The method is applicable to arbitrary channel configurations and maintains robust performance across high Nyquist zones. MATLAB simulations demonstrate that the proposed method achieves an SNDR of 67.54 dB, an SFDR of 94.09 dB, and an ENOB of 10.93 bits, with improvements of 11.67 dB (SNDR) and 32.74 dB (SFDR) being obtained. These results demonstrate the high accuracy, rapid convergence, and strong robustness of the proposed method, thereby confirming its suitability for arbitrary-channel and high-speed TIADC systems.

Author statement

Boyu Zhang: Methodology, Investigation, Writing – Original Draft, Writing – Review & Editing; Aang Hu: Supervision, Investigation,

Table 2
Comparisons with the state-of-the-art techniques.

References	[1] TCAS-I 2017[a]	[3] JSSC 2019	[2] IEEE Access 2020	[7] JSSC 2018	[24] TVLSI 2022	[25] JSSC 2025	This Work[a]
Channels	4	7/8-way(split)	16	16	2	4	**5**
SFDR [dB]	49.4/84.4	—/67.1	69.7/77.6	33.6/63.2	41.7/56.6	50.5/78.8	**61.35/94.09**
SNDR [dB]	46.4/60.1	—/54.2	40/48.5	45/50.6	40.3/48.5	49.5/59.3	**55.87/67.54**
Convergence Time	5 K	10K	16 K	6.5K	24.5K	10K	**10 K**
Filter	25-tap Polyphase FIR (+ Hilbert)	25-tap FIR (+Hilbert)	25-tap FIR (+Hilbert)	8 auxiliary delta-sampling ADCs(no FIR)	13-tap FIR	correlation (no FIR)	**3-tap FIR**
Extra Timing Ref. Channel	No	No	No	Yes	No	No	**No**
Calibration Type	Feedforward	Feedforward	Feedforward	Feedforward	Feedback	Feedforward	**Feedforward**
Arbitrary Channel	Yes	No	No	No	No	No	**Yes**
Nyquist Zone Applicability	1NZ-4NZ	1NZ-2NZ	1NZ-2NZ	1NZ	1NZ	1NZ	**1NZ-4NZ**

[a] Simulation result.

Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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